

内部言語の健全性定理と完全性定理

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概要

トポスの内部言語における健全性定理と完全性定理を証明する。

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$\mathcal{E}$ をトポス、 $\mathfrak{L}(\mathcal{E})$ を内部言語、 $\mathcal{M}(\mathcal{E})$ を標準モデル、 X を変数の有限集合とする。

文脈を増やすときは $\llbracket U \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})}$ というように、添字に変数を追加して書く。これは $\llbracket U \rrbracket_{X \cup \{y\}}^{\mathcal{M}(\mathcal{E})}$ と同じ意味。

1. 健全性定理

► 補題. 1.1 (文脈除去補題). $X \vdash U : \tau, x : \sigma, x \notin X$ ならば

$$\llbracket U \rrbracket_{X \cup \{x\}}^{\mathcal{M}(\mathcal{E})} = \llbracket U \rrbracket_X^{\mathcal{M}(\mathcal{E})} \circ \pi_{X \mathcal{M}(\mathcal{E})} : X^{\mathcal{M}(\mathcal{E})} \times \sigma^{\mathcal{M}(\mathcal{E})} \rightarrow X^{\mathcal{M}(\mathcal{E})} \rightarrow \tau^{\mathcal{M}(\mathcal{E})}$$

(文脈は適当に除去できる)

► 補題. 1.2 (代入補題). 項 $X \cup \{y : \tau\} \vdash U : \sigma, X \vdash V : \tau, y \notin X$ に対し

$$\llbracket U[y := V] \rrbracket_X^{\mathcal{M}(\mathcal{E})} = \llbracket U \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} \circ (1_{X \mathcal{M}(\mathcal{E})}, \llbracket V \rrbracket_X^{\mathcal{M}(\mathcal{E})})$$

(右辺は当に代入)

【証明】 項 U の構造に関して再帰的に証明する。

1. U が変数 $U = x : \sigma$ の時、

(a) $x = y$ なら $\sigma = \tau$ であって

$$\llbracket V \rrbracket_X^{\mathcal{M}(\mathcal{E})} = \llbracket x[y := V] \rrbracket_X^{\mathcal{M}(\mathcal{E})} = (X^{\mathcal{M}(\mathcal{E})} \rightarrow \sigma^{\mathcal{M}(\mathcal{E})})$$

は射影。また、 $\llbracket x \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} : X^{\mathcal{M}(\mathcal{E})} \times \sigma^{\mathcal{M}(\mathcal{E})} \rightarrow \sigma^{\mathcal{M}(\mathcal{E})}$ も射影である。従って、

$$\llbracket x \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} \circ (1_{X \mathcal{M}(\mathcal{E})}, \llbracket V \rrbracket_X^{\mathcal{M}(\mathcal{E})}) = \llbracket V \rrbracket_X^{\mathcal{M}(\mathcal{E})} = \llbracket x[y := V] \rrbracket_X^{\mathcal{M}(\mathcal{E})}$$

(b) $x \neq y$ なら $\llbracket x[y := V] \rrbracket_X^{\mathcal{M}(\mathcal{E})} = \llbracket x \rrbracket_X^{\mathcal{M}(\mathcal{E})} : X^{\mathcal{M}(\mathcal{E})} \rightarrow \sigma^{\mathcal{M}(\mathcal{E})}$ は射影である。

この時、射影 $\llbracket x \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} : X^{\mathcal{M}(\mathcal{E})} \times \tau^{\mathcal{M}(\mathcal{E})} \rightarrow \sigma^{\mathcal{M}(\mathcal{E})}$ は合成

$$X^{\mathcal{M}(\mathcal{E})} \times \tau^{\mathcal{M}(\mathcal{E})} \xrightarrow{\text{proj.}} X^{\mathcal{M}(\mathcal{E})} \xrightarrow{\llbracket x \rrbracket_X^{\mathcal{M}(\mathcal{E})}} \sigma^{\mathcal{M}(\mathcal{E})}$$

として書ける。よって

$$\llbracket x \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} \circ (1_{X \mathcal{M}(\mathcal{E})}, \llbracket V \rrbracket_X^{\mathcal{M}(\mathcal{E})}) = \llbracket x \rrbracket_X^{\mathcal{M}(\mathcal{E})} = \llbracket x[y := V] \rrbracket_X^{\mathcal{M}(\mathcal{E})}$$

2. $U = *$ の時は右辺も左辺も共に一意な射 $X^{\mathcal{M}(\mathcal{E})} \rightarrow 1$ になる。
 3. $X, y^v \vdash U : \sigma, X, y^v \vdash V : \tau, X \vdash W : v, y \notin X$ のとき

$$\begin{aligned}
& \llbracket (U, V)[y := W] \rrbracket_X^{\mathcal{M}(\mathcal{E})} \\
&= \llbracket (U[y := W], V[y := W]) \rrbracket_X^{\mathcal{M}(\mathcal{E})} \quad (\text{代入した}) \\
&= (\llbracket U[y := W] \rrbracket_X^{\mathcal{M}(\mathcal{E})}, \llbracket V[y := W] \rrbracket_X^{\mathcal{M}(\mathcal{E})}) \quad (\text{解釈の定義}) \\
&= (\llbracket U \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} \circ (1, \llbracket W \rrbracket_X^{\mathcal{M}(\mathcal{E})}), \llbracket V \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} \circ (1, \llbracket W \rrbracket_X^{\mathcal{M}(\mathcal{E})})) \quad (\text{再帰の仮定}) \\
&= (\llbracket U \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})}, \llbracket V \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})}) \circ (1, \llbracket W \rrbracket_X^{\mathcal{M}(\mathcal{E})}) \quad (\text{直積の性質}) \\
&= \llbracket (U, V) \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} \circ (1, \llbracket W \rrbracket_X^{\mathcal{M}(\mathcal{E})}) \quad (\text{解釈の定義})
\end{aligned}$$

4. $X, y^v \vdash U : \sigma \times \tau, X \vdash V : v, y \notin X$ のとき、

$$\begin{aligned}
& \llbracket \pi_i(U)[y := V] \rrbracket_X^{\mathcal{M}(\mathcal{E})} \\
&= \llbracket \pi_i(U[y := V]) \rrbracket_X^{\mathcal{M}(\mathcal{E})} \quad (\text{代入した}) \\
&= \pi_i \circ \llbracket U[y := V] \rrbracket_X^{\mathcal{M}(\mathcal{E})} \quad (\text{解釈の定義}) \\
&= \pi_i \circ \llbracket U \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} \circ (1, \llbracket V \rrbracket_X^{\mathcal{M}(\mathcal{E})}) \quad (\text{再帰の仮定}) \\
&= \llbracket \pi_i(U) \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} \circ (1, \llbracket V \rrbracket_X^{\mathcal{M}(\mathcal{E})}) \quad (\text{解釈の定義})
\end{aligned}$$

5. 関数記号 $f : \sigma_1, \dots, \sigma_n \rightarrow \tau$ と $X, y^v \vdash U_i : \sigma_i, X \vdash V : v$ に対し

$$\begin{aligned}
& \llbracket f((U_i)_{i=1}^n)[y := V] \rrbracket_X^{\mathcal{M}(\mathcal{E})} \\
&= \llbracket f((U_i[y := V])_{i=1}^n) \rrbracket_X^{\mathcal{M}(\mathcal{E})} \quad \text{代入した} \\
&= f^{\mathcal{M}(\mathcal{E})} \circ (\llbracket U_i[y := V] \rrbracket_X^{\mathcal{M}(\mathcal{E})})_{i=1}^n \quad (\text{解釈の定義}) \\
&= f^{\mathcal{M}(\mathcal{E})} \circ (\llbracket U_i \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} \circ (1, \llbracket V \rrbracket_X^{\mathcal{M}(\mathcal{E})}))_{i=1}^n \quad (\text{再帰の仮定}) \\
&= f^{\mathcal{M}(\mathcal{E})} \circ (\llbracket U_i \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})})_{i=1}^n \circ (1, \llbracket V \rrbracket_X^{\mathcal{M}(\mathcal{E})}) \quad (\text{直積の性質}) \\
&= \llbracket f((U_i)_{i=1}^n) \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} \circ (1, \llbracket V \rrbracket_X^{\mathcal{M}(\mathcal{E})}) \quad \text{解釈の定義}
\end{aligned}$$

6. $X, y^v \vdash U : \sigma, X, y^v \vdash V : \sigma, X \vdash W : v, y \notin X$ に対し

$$\begin{aligned}
& \llbracket (U =_\sigma V)[y := W] \rrbracket_X^{\mathcal{M}(\mathcal{E})} \\
&= \llbracket (U[y := W] =_\sigma V[y := W]) \rrbracket_X^{\mathcal{M}(\mathcal{E})} \\
&= (=_{\sigma^{\mathcal{M}(\mathcal{E})}}) \circ (\llbracket U[y := W] \rrbracket_X^{\mathcal{M}(\mathcal{E})}, \llbracket V[y := W] \rrbracket_X^{\mathcal{M}(\mathcal{E})}) \\
&= (=_{\sigma^{\mathcal{M}(\mathcal{E})}}) \circ (\llbracket U \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})}, \llbracket V \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})}) \circ (1, \llbracket W \rrbracket_X^{\mathcal{M}(\mathcal{E})})
\end{aligned}$$

$$= \llbracket U =_\sigma V \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} \circ (1, \llbracket W \rrbracket_X^{\mathcal{M}(\mathcal{E})})$$

7. $X, y^v \vdash U : \sigma, X, y^v \vdash V : P(\sigma), X \vdash W : v, y \notin X$ に対し

$$\begin{aligned} & \llbracket (U \in_\sigma V)[y := W] \rrbracket_X^{\mathcal{M}(\mathcal{E})} \\ &= \llbracket (U[y := W] \in_\sigma V[y := W]) \rrbracket_X^{\mathcal{M}(\mathcal{E})} \\ &= (\in_{\sigma \mathcal{M}(\mathcal{E})}) \circ (\llbracket U[y := W] \rrbracket_X^{\mathcal{M}(\mathcal{E})}, \llbracket V[y := W] \rrbracket_X^{\mathcal{M}(\mathcal{E})}) \\ &= (\in_{\sigma \mathcal{M}(\mathcal{E})}) \circ (\llbracket U \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})}, \llbracket V \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})}) \circ (1, \llbracket W \rrbracket_X^{\mathcal{M}(\mathcal{E})}) \\ &= \llbracket U \in_\sigma V \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} \circ (1, \llbracket W \rrbracket_X^{\mathcal{M}(\mathcal{E})}) \end{aligned}$$

8. $X, x^\sigma, y^\tau \vdash \varphi : \Omega, X \vdash U : \tau, x \notin X, y \notin X, x \neq y$ として

$$\llbracket \{x : \sigma \mid \varphi\}[y := U] \rrbracket_X^{\mathcal{M}(\mathcal{E})}$$

を考える。

$$\begin{aligned} & \llbracket \{x : \sigma \mid \varphi\}[y := U] \rrbracket_X^{\mathcal{M}(\mathcal{E})} \\ &= \llbracket \{x : \sigma \mid \varphi[y := U]\} \rrbracket_X^{\mathcal{M}(\mathcal{E})} \\ &\stackrel{1:1}{\leftrightarrow} \llbracket \varphi[y := U] \rrbracket_{X,x}^{\mathcal{M}(\mathcal{E})} \\ &= \llbracket \varphi \rrbracket_{X,x,y}^{\mathcal{M}(\mathcal{E})} \circ (1, \llbracket U \rrbracket_{X,x}^{\mathcal{M}(\mathcal{E})}) \end{aligned}$$

今、 $\llbracket U \rrbracket_{X,x}^{\mathcal{M}(\mathcal{E})} = \llbracket U \rrbracket_X^{\mathcal{M}(\mathcal{E})} \circ \pi_{X \mathcal{M}(\mathcal{E})}$ なので、以下は可換：

$$\begin{array}{ccc} \mathcal{E}(X^{\mathcal{M}(\mathcal{E})} \times \sigma^{\mathcal{M}(\mathcal{E})} \times \tau^{\mathcal{M}(\mathcal{E})}, \Omega) & \xrightarrow{\sim} & \mathcal{E}(X^{\mathcal{M}(\mathcal{E})} \times \tau^{\mathcal{M}(\mathcal{E})}, \Omega) \\ \downarrow - \circ (1, \llbracket U \rrbracket_{X,x}^{\mathcal{M}(\mathcal{E})}) & & \downarrow - \circ (1, \llbracket U \rrbracket_X^{\mathcal{M}(\mathcal{E})}) \\ \mathcal{E}(X^{\mathcal{M}(\mathcal{E})} \times \sigma^{\mathcal{M}(\mathcal{E})}, \Omega) & \xrightarrow{\sim} & \mathcal{E}(X^{\mathcal{M}(\mathcal{E})}, \Omega) \end{array}$$

よって $\llbracket \varphi \rrbracket_{X,x,y}^{\mathcal{M}(\mathcal{E})} \circ (1, \llbracket U \rrbracket_{X,x}^{\mathcal{M}(\mathcal{E})})$ は $\llbracket \{x : \sigma \mid \varphi\} \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} \circ (1, \llbracket U \rrbracket_X^{\mathcal{M}(\mathcal{E})})$ と対応する。

従って、

$$\llbracket \{x : \sigma \mid \varphi\}[y := U] \rrbracket_X^{\mathcal{M}(\mathcal{E})} = \llbracket \{x : \sigma \mid \varphi\} \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} \circ (1, \llbracket U \rrbracket_X^{\mathcal{M}(\mathcal{E})})$$



► 補題. 1.3. $\mathcal{M}(\mathcal{E})$ は $\mathfrak{T}$ の基本公理を満たす。

【証明】

1. (Unity) $\vdash_{x:1} x =_1 *$:

$x:1$ の時、 $\llbracket x \rrbracket_{x:1}^{\mathcal{M}(\mathcal{E})} = (1 \rightarrow 1)$ なので

$$\llbracket x =_1 * \rrbracket_x^{\mathcal{M}(\mathcal{E})} = (=_1) \circ (1, 1) = \text{true}$$

2. (Equality0) $\vdash_X U =_\sigma U$:

$$\begin{aligned} \llbracket U =_\sigma U \rrbracket_X^{\mathcal{M}(\mathcal{E})} &= (=_{\sigma \mathcal{M}(\mathcal{E})}) \circ (\llbracket U \rrbracket_X^{\mathcal{M}(\mathcal{E})}, \llbracket U \rrbracket_X^{\mathcal{M}(\mathcal{E})}) \\ &= (=_{\sigma \mathcal{M}(\mathcal{E})}) \circ \Delta_{\sigma \mathcal{M}(\mathcal{E})} \circ \llbracket U \rrbracket_X^{\mathcal{M}(\mathcal{E})} \\ &= \text{true} \end{aligned}$$

3. (Equality1)



► 補題. 1.4. 各推論規則の前提が $\mathcal{M}(\mathcal{E})$ で満たされるなら、その結論も満たされる。

► 定理. 1.5. シークウェント $\Gamma \vdash_X \varphi$ が $\mathfrak{T}$ で導出可能なら、 $\mathcal{M}(\mathcal{E})$ は $\Gamma \vdash_X \varphi$ を満たす。

2. 完全性定理

► 定理. 2.1. $\mathcal{M}(\mathcal{E})$ が $\Gamma \vdash_X \varphi$ を満たすなら、 $\Gamma \vdash_X \varphi$ は $\mathfrak{T}$ で導出可能である。